

KAUNG HTET CHO - AI ENGINEER

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SUMMARY

AI Engineer with hands-on experience across computer vision, audio classification, medical imaging, and generative AI – building systems that go from research to production. As a research assistant, I have deployed real-time inference pipelines, architected distributed field hardware systems, and conducted clinical deep learning research. A background in factory automation gives me a systems-level reliability mindset. Currently completing a Master's in Mechatronics and AI at AIT, available to start immediately.

WORK EXPERIENCE

AI Center (Asian Institute of Technology, Bangkok, Thailand)

May 2024 - Present

Position: **Research Assistant (AI Engineer)**

Worked on multiple applied AI research and deployment projects including surveillance analytics, biodiversity monitoring, and medical imaging.

Project: Area Surveillance – Real-Time Speed Detection System

- Designed and deployed a real-time vehicle speed detection pipeline using NVIDIA DeepStream, achieving ~95% detection accuracy across dual RTSP camera streams
- Engineered computer vision algorithms to calculate vehicle speed from multi-camera inputs, replacing fully manual monitoring workflows
- Built a backend system with database storage and web-based analytics dashboard for live surveillance reporting, using RabbitMQ for internal pipeline messaging and Redis pub/sub to deliver real-time results to external web clients
- Implemented CI/CD pipelines (GitLab CI) to automate model testing and deployment, reducing manual deployment

Technologies: Python, NVIDIA DeepStream, Docker, CI/CD, RTSP Streaming, Redis, RabbitMQ

Project: Biodiversity Monitoring – Bird and Insect Sound AI Pipeline

- Architected end-to-end AI system from field hardware to cloud inference, ingesting continuous 24/7 audio streams from 25 field devices across 12 monitoring sites
- Built automated audio ingestion pipelines processing 5-second audio chunks at real-time throughput, handling ~10 minutes of overlapping wildlife audio per inference cycle
- Trained deep learning models for bird and insect sound classification, achieving 93% accuracy across 52 species
- Deployed daily analytics pipeline for bird and insect population scoring, delivering results via REST APIs and JSON feeds to mobile and web platforms
- Set up monitoring dashboards using Grafana and Cacti to track pipeline health across all 25 field devices

Technologies: Python, PyTorch, Docker, Linux, REST API, Grafana, Cacti

Project: Leukemia Cell Detection – Medical AI Research

- Developed a two-stage deep learning pipeline for localizing and classifying leukemic vs. non-leukemic white blood cells from microscopic images
- Finetuned self-supervised learning models (DINO, DINO-Bloom) to improve performance under limited labeled data conditions, achieving 88.87% accuracy with 92%+ precision on leukemic class
- Configured CVAT annotation platform to support large-scale medical image labeling across 21,000 images
- Optimized model for cross-domain generalization across multiple medical datasets, improving robustness by 5%

Technologies: PyTorch, Transformers, Detectron2, CVAT

Project: AI Deployment, Monitoring & Security Infrastructure

- Built GitLab CI/CD test pipelines for multiple AI models in production surveillance systems, including synthetic test video case generation to validate model performance across diverse scenarios
- Visualized system monitoring dashboards using Grafana and Portainer to manage Docker containers and 100 camera systems from a single interface
- Secured hard disk storage on edge AI servers using LUKS full-disk encryption with TPM2 hardware-based auto-unlock, achieving zero unauthorized access incidents

Technologies: Docker, Grafana, Linux, GitLab CI/CD, TPM2, LUKS

JFE Meranti Myanmar - Yangon, Myanmar

Sep 2019 - Sep 2022

Position: [Operation and Maintenance Engineer](#)

- Performed installation & commissioning of precision machines (EDT, Heavy grinder, Semi-auto Lathe, De-chocking machines) cooperating with multinational workers in a 6-month duration
- Moderated preventive and break down maintenance of hydraulic and pneumatic systems, sensors, actuators and electrical systems of CNC machines and production line

Delta Electronics - Yangon, Myanmar

May 2019 - Aug 2019

Position: [Process Engineer](#)

- Led the IE/PE team; responsible for monitoring production line and support 4M methods for pilot-line operation
- Collaborated with QA/QC and production depts performing Root Cause Analysis for delayed WIP
- Designed jigs and fixtures designs applying AutoCAD and GD&T

EDUCATION

Master of Engineering (Mechatronics and Artificial Intelligence)

Aug 2023 - May 2026

Asian Institute of Technology, Bangkok, Thailand

Thesis : "A transformer Based Two-Stage Framework for Acute Leukemia Cell Detection and Classification"

Projects: (All the source codes can be found at my GitHub)

- Serverless face recognition application using AWS services
- Robot arm control using Deep Reinforcement Learning
- Football Match Analyzer (computer vision)
- Medical QA chatbot with LangChain (LLM)
- ML-based carbon emission forecasting for Thailand

Bachelor of Engineering (Mechanical Precision and Automation)

Dec 2012 - Sep 2019

University of Technology (Yatanarpon Cybercity), Mandalay, Myanmar

Thesis : "Kinematics Analysis of 6 DoF Robotics Arm"

TECHNICAL SKILLS

- [Programming Languages](#): Python, C++
- [Web Technologies and Frameworks](#): React, HTML, CSS
- [Generative AI & LLM Systems](#): LangChain, LlamaIndex, Retrieval-Augmented Generation (RAG), Chroma, Qdrant, Agentic AI, Google ADK, Prompt Engineering, Context Engineering, Fine-Tuning (LoRA, QLoRA), Local Model Serving (llama.cpp, Ollama), Cloud Model Serving (OpenAI, Gemini APIs), Hugging Face
- [Cloud Computing](#): AWS, Google Cloud Platform
- [Edge and Embedded](#): NVIDIA Jetson, Raspberry Pi, IOT devices
- [Database](#): MySQL, MongoDB, Neo4j
- [MLOps & Deployment](#): CI/CD (GitLab CI, GitHub Actions), Kubernetes, Docker, KubeFlow, MLFlow, Grafana, Cacti
- [Machine Learning and Deep Learning](#): Supervised Learning, Unsupervised Learning, Self-Supervised Learning, Neural Networks, Computer Vision, Natural Language Processing (NLP), Audio Processing, Object Detection (YOLO, Faster R-CNN, Detectron2, DETR), Vision-Language Model, SAM, DINO
- [AI Frameworks and tools](#): PyTorch, TensorFlow, Detectron2
- [Others](#): Redis, RabbitMQ, REST API, Version Control with Git, Linux, Postman, SDLC

CERTIFICATION

- Introduction to Kubernetes - The Linux Foundation (2026)
- AI/ML Toolkits with Kubeflow - The Linux Foundation (2026)
- Introduction to GitOps - The Linux Foundation (2026)
- Secure AI/ML-Driven Software Development - The Linux Foundation (2026)
- Vector Search - Qdrant (2026)
- MCP for Beginners - KodeKloud (2026)
- Generative AI LLMs Associate Certification - Nvidia (2026)
- Fundamental of MLOps - KodeKloud (2026)